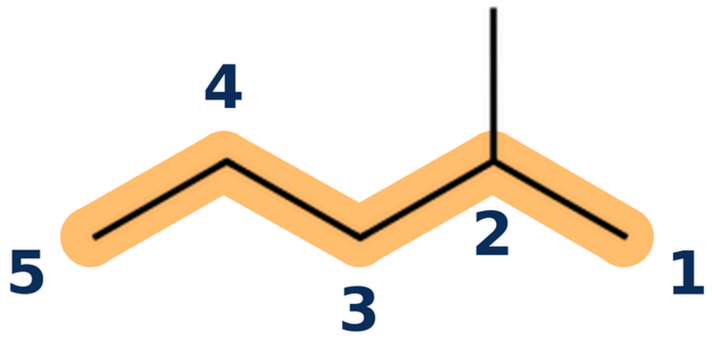
A01 **FOUNDATION**

Molecular formula (counted): C6H14

1 · NUMBERED PARENT



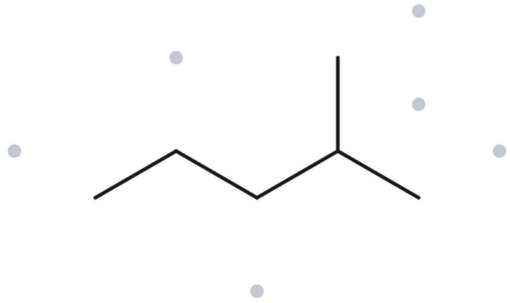
2 · NAMING TRAIL

Parent: Longest chain = 5 C → pentane. The drawn vertical stub is a methyl branch, not part of the chain.
Numbering: Left → right: methyl on C4. Right → left: methyl on C2. 2 < 4 → number from the right.
Substituents: methyl on C2 → 2-methyl.

3 · MY NAME

2-methylpentane

4 · THE ALKANE CHECK



2-methylpentane

Returned name: 2-methylpentane

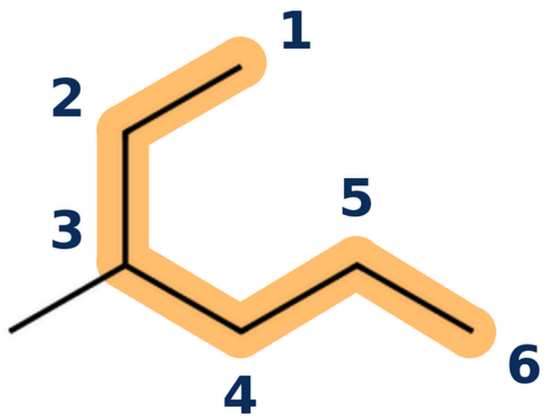
■✓ Match □✗ Mismatch (state the correct name + the rule)

✓ longest chain: 5 C ✓ locants: {2} < {4} ✓ order: methyl only ✓ punctuation, no spaces ✓ formula: C6H14

A02 **FOUNDATION**

Molecular formula (counted): C7H16

1 · NUMBERED PARENT



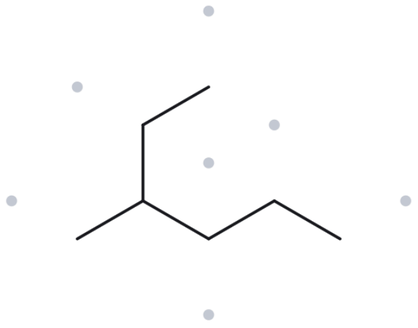
2 · NAMING TRAIL

Parent: Longest chain = 6 C → hexane. It bends: up the drawn ethyl-looking arm, through the branch carbon, then along the 3-C run at right. The horizontal run is only 5 C.
Numbering: From the top end: methyl on C3. From the bottom-right end: methyl on C4. 3 < 4 → number from the top end.
Substituents: methyl on C3 → 3-methyl.

3 · MY NAME

3-methylhexane

4 · THE ALKANE CHECK



3-methylhexane

Returned name: 3-methylhexane

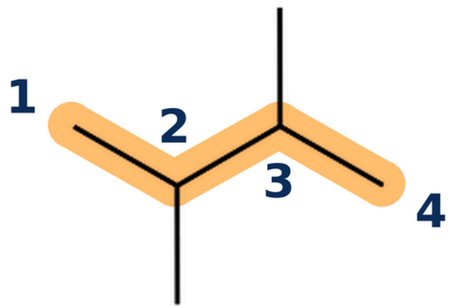
■✓ Match □✗ Mismatch (state the correct name + the rule)

✓ longest chain: 6 C | ✓ locants: {3} < {4} | ✓ order: methyl only | ✓ punctuation, no spaces | ✓ formula: C7H16

A03 **FOUNDATION**

Molecular formula (counted): C6H14

1 · NUMBERED PARENT



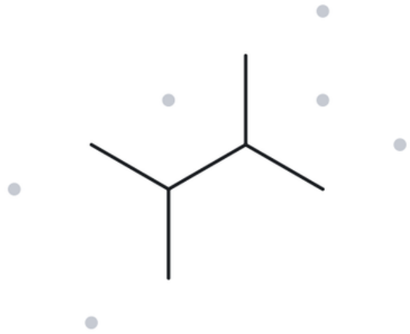
2 · NAMING TRAIL

Parent: Longest chain = 4 C → butane. No 5-C path exists; the several 4-C paths all give the same name.
Numbering: Left → right: methyls on C2 and C3 → {2,3}. Right → left: {2,3}. Tie; either direction gives 2,3.
Substituents: methyl on C2 and methyl on C3 → 2,3-dimethyl (identical groups: di-, one locant each).

3 · MY NAME

2,3-dimethylbutane

4 · THE ALKANE CHECK



2,3-dimethylbutane

Returned name: 2,3-dimethylbutane

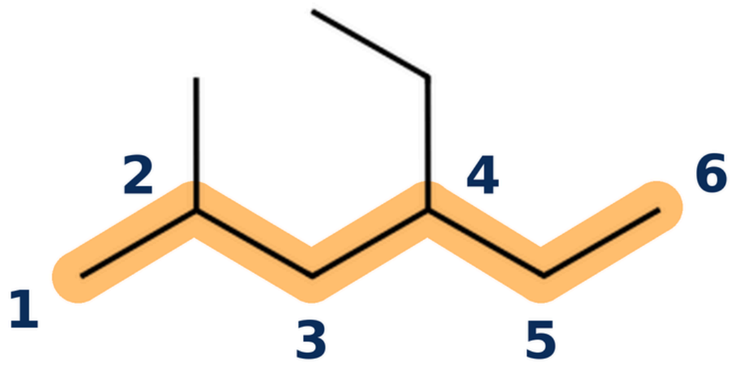
■✓ Match □ ✗ Mismatch (state the correct name + the rule)

✓ longest chain: 4 C | ✓ locants: {2,3} tie | ✓ order: methyl ×2 (di-) | ✓ punctuation, no spaces | ✓ formula: C6H14

A04 DEVELOPING

Molecular formula (counted): C9H20

1 · NUMBERED PARENT



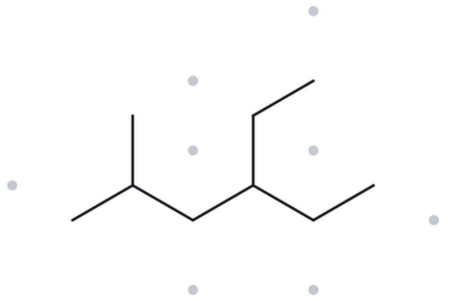
2 · NAMING TRAIL

Parent: Longest chain = 6 C → hexane. Four 6-C paths exist (via the methyl or ethyl arm); all give the same name.
Numbering: Left → right: methyl C2, ethyl C4 → {2,4}. Right → left: ethyl C3, methyl C5 → {3,5}. 2 < 3 at the first point of difference → number from the left.
Substituents: Alphabetical, not locant order: ethyl (C4), then methyl (C2) → 4-ethyl-2-methyl.

3 · MY NAME

4-ethyl-2-methylhexane

4 · THE ALKANE CHECK



4-ethyl-2-methylhexane

Returned name: 4-ethyl-2-methylhexane

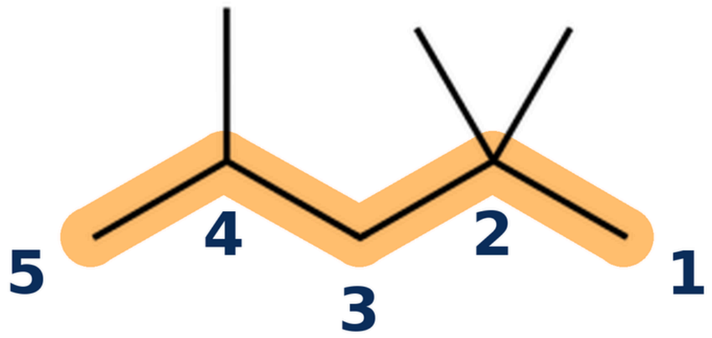
■✓ Match □✗ Mismatch (state the correct name + the rule)

✓ longest chain: 6 C | ✓ locants: {2,4} < {3,5} | ✓ order: ethyl, methyl | ✓ punctuation, no spaces | ✓ formula: C9H20

A05 DEVELOPING

Molecular formula (counted): C8H18

1 · NUMBERED PARENT



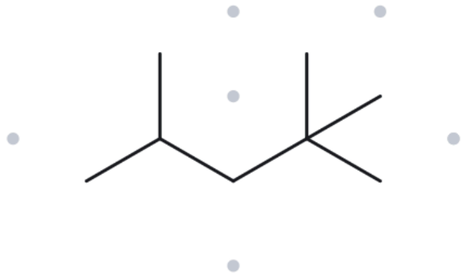
2 · NAMING TRAIL

Parent: Longest chain = 5 C → pentane. The two methyls on the quaternary carbon do not lengthen it.
Numbering: Right → left: methyls on C2, C2, C4 → {2,2,4}. Left → right: {2,4,4}. First point of difference: 2 = 2, then 2 < 4 → number from the right.
Substituents: methyl on C2, C2 and C4 → 2,2,4-trimethyl (one locant per group, repeated).

3 · MY NAME

2,2,4-trimethylpentane

4 · THE ALKANE CHECK



2,2,4-trimethylpentane

Returned name: 2,2,4-trimethylpentane

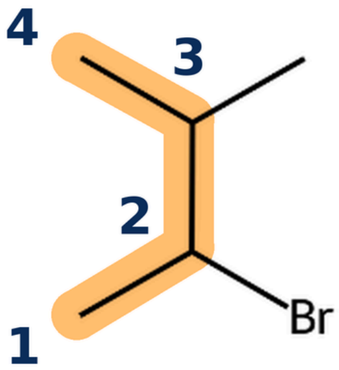
■✓ Match □✗ Mismatch (state the correct name + the rule)

✓ longest chain: 5 C ✓ locants: {2,2,4} < {2,4,4} ✓ order: methyl ×3 (tri-) ✓ punctuation, no spaces ✓ formula: C8H18

A06 DEVELOPING

Molecular formula (counted): C5H11Br

1 · NUMBERED PARENT



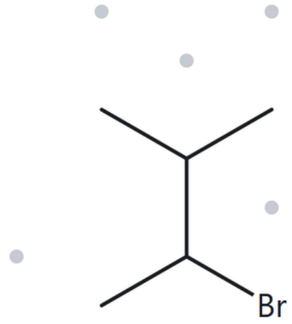
2 · NAMING TRAIL

Parent: Longest chain = 4 C → butane. Bromine is a substituent (prefix bromo), not part of the chain.
Numbering: Bottom → top: bromo C2, methyl C3 → {2,3}. Top → bottom: methyl C2, bromo C3 → {2,3}. Tie → alphabetical order decides: bromo comes before methyl, so bromo gets the lower number.
Substituents: bromo on C2, methyl on C3 → 2-bromo-3-methyl.

3 · MY NAME

2-bromo-3-methylbutane

4 · THE ALKANE CHECK



2-bromo-3-methylbutane

Returned name: 2-bromo-3-methylbutane

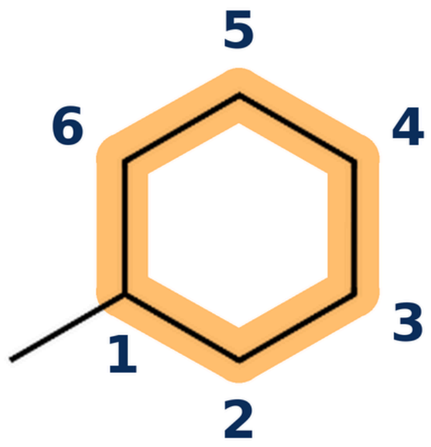
■✓ Match □ ✗ Mismatch (state the correct name + the rule)

✓ longest chain: 4 C ✓ locants: {2,3} tie ✓ order: bromo, methyl ✓ punctuation, no spaces ✓ formula: C5H11Br

A07 DEVELOPING

Molecular formula (counted): C7H14

1 · NUMBERED PARENT



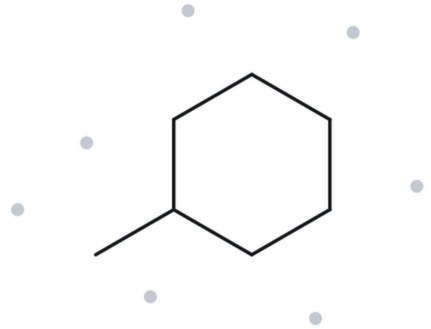
2 · NAMING TRAIL

Parent: A ring is present, so the ring is the parent: 6 C ring → cyclohexane.
Numbering: One substituent: it takes C1 by default, so no locant is written (methylcyclohexane, not 1-methylcyclohexane).
Substituents: methyl on C1 → methyl.

3 · MY NAME

methylcyclohexane

4 · THE ALKANE CHECK



methylcyclohexane

Returned name: methylcyclohexane

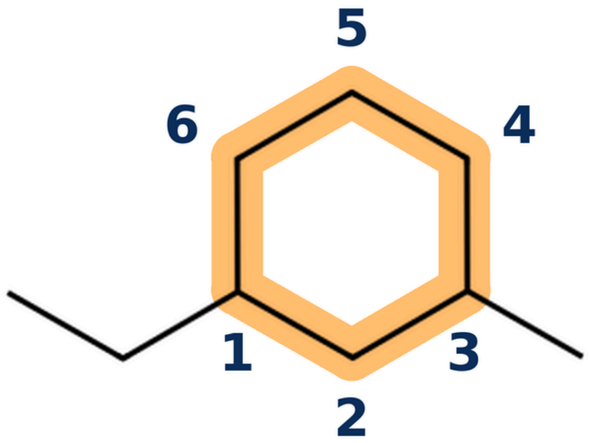
■✓ Match □✗ Mismatch (state the correct name + the rule)

✓ parent ring: 6 C ✓ locant: none needed ✓ order: methyl only ✓ punctuation, no spaces ✓ formula: C7H14

A 0 8 **CHALLENGE**

Molecular formula (counted): C9H18

1 · NUMBERED PARENT



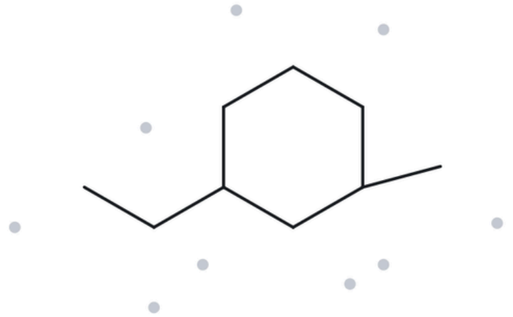
2 · NAMING TRAIL

Parent: Ring present → the ring is the parent: 6 C → cyclohexane. Ethyl and methyl are substituents.
Numbering: Going the short way round, the substituents are on carbons 1 and 3, from either one → {1,3} both ways. Tie → alphabetical order decides: ethyl gets C1.
Substituents: ethyl on C1, methyl on C3 → 1-ethyl-3-methyl.

3 · MY NAME

1-ethyl-3-methylcyclohexane

4 · THE ALKANE CHECK



1-ethyl-3-methylcyclohexane

Returned name: 1-ethyl-3-methylcyclohexane

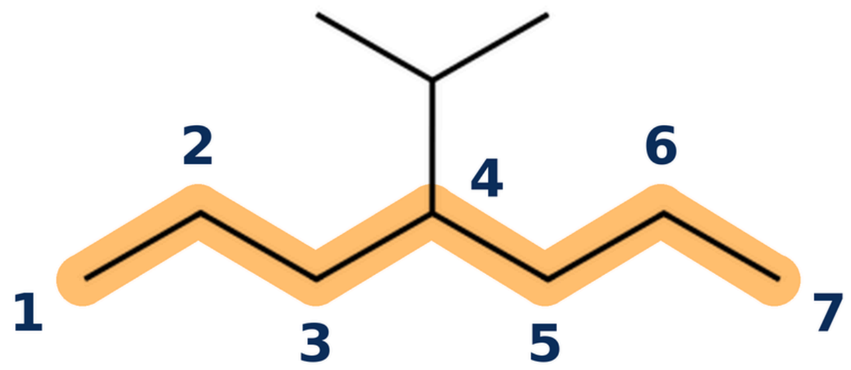
■✓ Match □ ✗ Mismatch (state the correct name + the rule)

✓ parent ring: 6 C ✓ locants: {1,3} tie ✓ order: ethyl, methyl ✓ punctuation, no spaces ✓ formula: C9H18

A 0 9 **CHALLENGE**

Molecular formula (counted): C10H22

1 · NUMBERED PARENT



2 · NAMING TRAIL

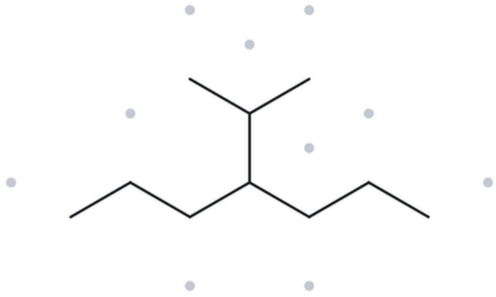
Parent: Longest chain = 7 C → heptane (the horizontal run). The path through the branch is only 6 C, so the branched group is a substituent.
Numbering: Both directions put the branch on C4 → {4} either way; number from the left.
Substituents: propan-2-yl (isopropyl), (CH3)2CH–, on C4 → 4-isopropyl.

3 · MY NAME

4-isopropylheptane

Also accepted: 4-(1-methylethyl)heptane · 4-(propan-2-yl)heptane

4 · THE ALKANE CHECK



2-methyl-3-propylhexane

Returned name: 2-methyl-3-propylhexane

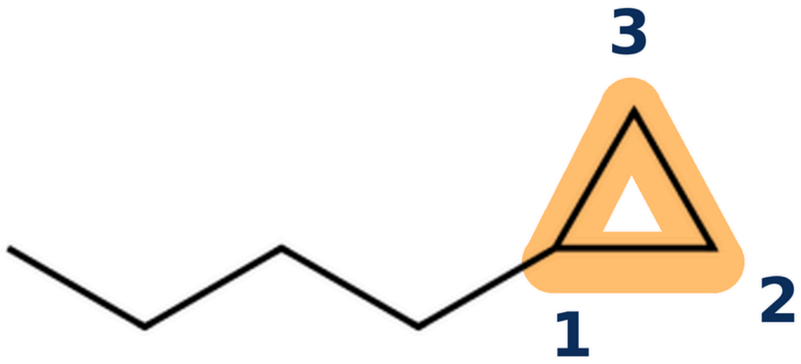
☐ ✓ Match ☐ ✗ Mismatch (s **(4-isopropylheptane, parent must be the longest chain (7 C, not 6 C))**

✓ longest chain: 7 C **✓ locant: {4} both ways** **✓ order: isopropyl only** **✓ punctuation, no spaces** **✓ formula: C10H22**

A10 CHALLENGE

Molecular formula (counted): C7H14

1 · NUMBERED PARENT



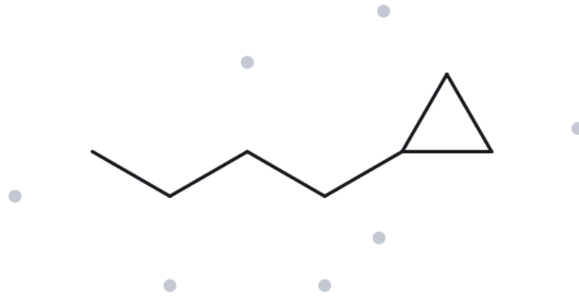
2 · NAMING TRAIL

Parent: Ring present → the ring is the parent: cyclopropane (3 C), even though the attached chain (4 C) is longer (ring-first convention of this activity, IUPAC 2013).
Numbering: One substituent: it takes C1 by default, so no locant is written.
Substituents: butyl (4-C chain) on C1 → butyl.

3 · MY NAME

butylcyclopropane

4 · THE ALKANE CHECK



butylcyclopropane

Returned name: butylcyclopropane

■✓ Match □ ✗ Mismatch (state the correct name + the rule)

✓ parent ring: 3 C | ✓ locant: none needed | ✓ order: butyl only | ✓ punctuation, no spaces | ✓ formula: C7H14

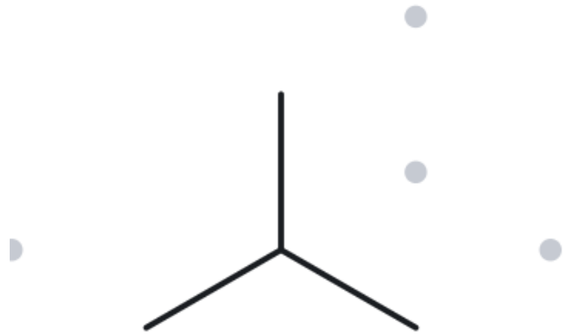
B01 FOUNDATION

Given name: 2-methylpropane

1 · NAME DECODED

Parent: propane → 3-C chain
Substituents: methyl on C2
Expected formula: 3 + 1 = 4 C; open chain → C4H10

2 · BOND-LINE STRUCTURE (THE ALKANE)



2-methylpropane

3 · FORMULA CHECK

Counted from the drawing: C4H10
Open chain: CnH2n+2 = C4H10 ✓

4 · BACK-CHECK

Returned by The Alkane: 2-methylpropane
■ Same as the given name
□ Different. Explanation: _____

✓ parent length: 3 C | ✓ positions: methyl on C2 | ✓ formula: C4H10 | □ back-check recorded

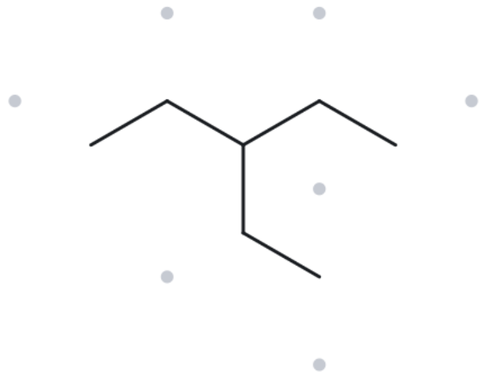
B02 FOUNDATION

Given name: 3-ethylpentane

1 · NAME DECODED

Parent: pentane → 5-C chain
Substituents: ethyl on C3
Expected formula: 5 + 2 = 7 C; open chain → C7H16

2 · BOND-LINE STRUCTURE (THE ALKANE)



3-ethylpentane

3 · FORMULA CHECK

Counted from the drawing: C7H16
Open chain: CnH2n+2 = C7H16 ✓

4 · BACK-CHECK

Returned by The Alkane: **3-ethylpentane**
■ Same as the given name
□ Different. Explanation: _____

✓ parent length: 5 C | ✓ positions: ethyl on C3 | ✓ formula: C7H16 | □ back-check recorded

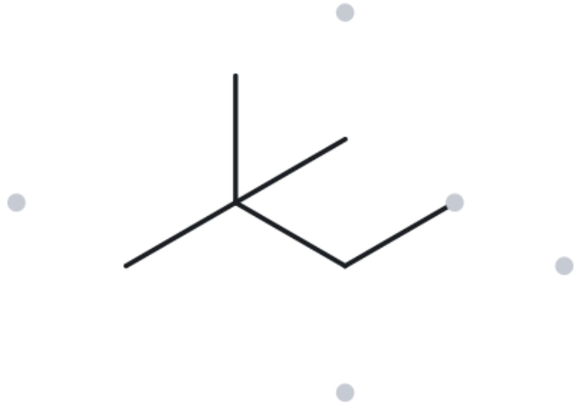
B03 FOUNDATION

Given name: 2,2-dimethylbutane

1 · NAME DECODED

Parent: butane → 4-C chain
Substituents: methyl on C2, methyl on C2
Expected formula: 4 + 2 = 6 C; open chain → C6H14

2 · BOND-LINE STRUCTURE (THE ALKANE)



2,2-dimethylbutane

3 · FORMULA CHECK

Counted from the drawing: C6H14
Open chain: CnH2n+2 = C6H14 ✓

4 · BACK-CHECK

Returned by The Alkane: **2,2-dimethylbutane**_____
■ Same as the given name
□ Different. Explanation: _____

✓ parent length: 4 C | ✓ positions: methyl on C2 | ✓ formula: C6H14 | □ back-check recorded

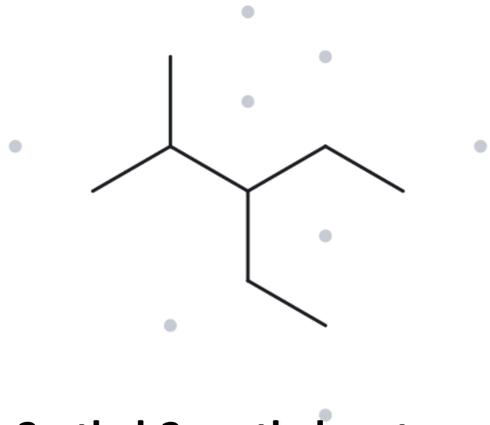
B04 DEVELOPING

Given name: 3-ethyl-2-methylpentane

1 · NAME DECODED

Parent: pentane → 5-C chain
Substituents: ethyl on C3, methyl on C2
Expected formula: 5 + 2 + 1 = 8 C; open chain → C8H18

2 · BOND-LINE STRUCTURE (THE ALKANE)



3-ethyl-2-methylpentane

3 · FORMULA CHECK

Counted from the drawing: C8H18
Open chain: CnH2n+2 = C8H18 ✓

4 · BACK-CHECK

Returned by The Alkane: **3-ethyl-2-methylpentane**
■ Same as the given name
□ Different. Explanation: _____

✓ parent length: 5 C | ✓ positions: ethyl on C3 | ✓ formula: C8H18 | □ back-check recorded

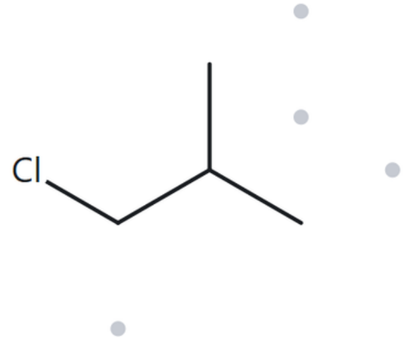
B05 DEVELOPING

Given name: 1-chloro-2-methylpropane

1 · NAME DECODED

Parent: propane → 3-C chain
Substituents: chloro on C1, methyl on C2
Expected formula: 3 + 1 = 4 C; C4H10, Cl replaces one H → C4H9Cl

2 · BOND-LINE STRUCTURE (THE ALKANE)



1-chloro-2-methylpropane

3 · FORMULA CHECK

Counted from the drawing: C4H9Cl
Open chain: CnH2n+2 = C4H10; minus one H for Cl = C4H9Cl ✓

4 · BACK-CHECK

Returned by The Alkane: 1-chloro-2-methylpropane
■ Same as the given name
□ Different. Explanation: _____

✓ parent length: 3 C | ✓ positions: chloro on C1 | ✓ formula: C4H9Cl | □ back-check recorded

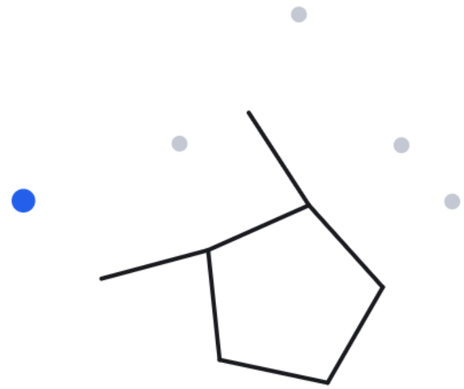
B06 DEVELOPING

Given name: 1,2-dimethylcyclopentane

1 · NAME DECODED

Parent: cyclopentane → 5-C ring
Substituents: methyl on C1, methyl on C2
Expected formula: 5 + 2 = 7 C; one ring → C7H14

2 · BOND-LINE STRUCTURE (THE ALKANE)



1,2-dimethylcyclopentane

3 · FORMULA CHECK

Counted from the drawing: C7H14
One ring: CnH2n = C7H14 ✓

4 · BACK-CHECK

Returned by The Alkane: 1,2-dimethylcyclopentane
■ Same as the given name
□ Different. Explanation: _____

✓ parent length: 5 C | ✓ positions: methyl on C1 | ✓ formula: C7H14 | □ back-check recorded

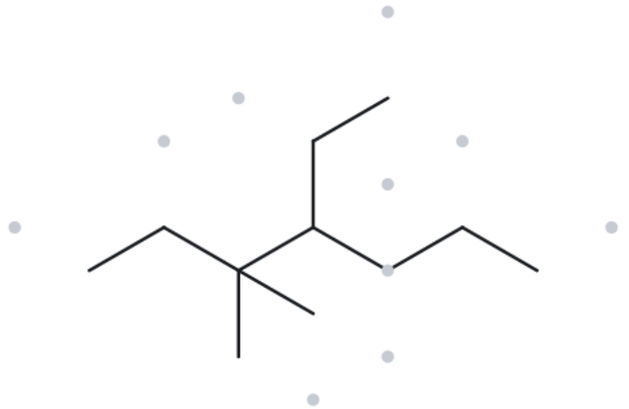
B 0 7 **DEVELOPING**

Given name: 4-ethyl-3,3-dimethylheptane

1 · NAME DECODED

Parent: heptane → 7-C chain
Substituents: methyl on C3 (×2), ethyl on C4
Expected formula: 7 + 2 + 2 = 11 C; open chain → C11H24

2 · BOND-LINE STRUCTURE (THE ALKANE)



4-ethyl-3,3-dimethylheptane

3 · FORMULA CHECK

Counted from the drawing: C11H24
Open chain: CnH2n+2 = C11H24 ✓

4 · BACK-CHECK

Returned by The Alkane: **4-ethyl-3,3-dimethylheptane**
■ Same as the given name
□ Different. Explanation: _____

✓ **parent length: 7 C** ✓ **positions: methyl on C3 (×2)** ✓ **formula: C11H24** □ **back-check recorded**

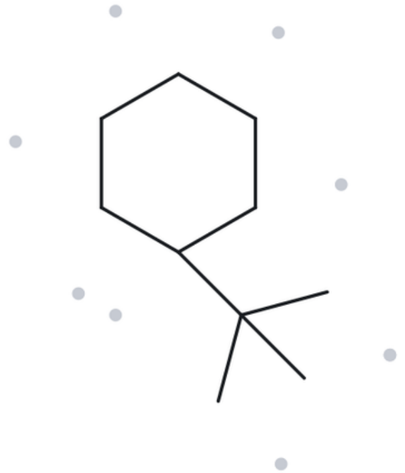
B 0 8 CHALLENGE

Given name: tert-butylcyclohexane

1 · NAME DECODED

Parent: cyclohexane → 6-C ring
Substituents: tert-butyl, (CH3)3C– (4 C), on C1
Expected formula: 6 + 4 = 10 C; one ring → C10H20

2 · BOND-LINE STRUCTURE (THE ALKANE)



tert-butylcyclohexane

3 · FORMULA CHECK

Counted from the drawing: C10H20
One ring: CnH2n = C10H20 ✓

4 · BACK-CHECK

Returned by The Alkane: tert-butylcyclohexane
■ Same as the given name
□ Different. Explanation: _____

✓ parent length: 6 C | ✓ positions: tert-butyl | ✓ formula: C10H20 | □ back-check recorded

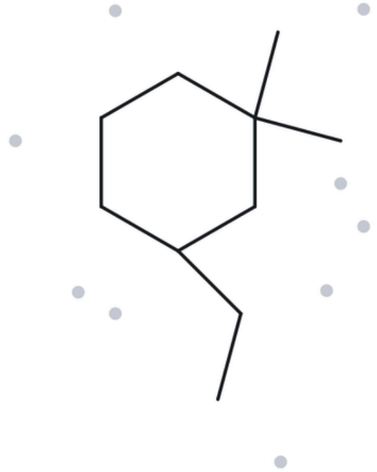
B 0 9 **CHALLENGE**

Given name: 3-ethyl-1,1-dimethylcyclohexane

1 · NAME DECODED

Parent: cyclohexane → 6-C ring
Substituents: methyl on C1 (×2), ethyl on C3
Expected formula: 6 + 2 + 2 = 10 C; one ring → C10H20

2 · BOND-LINE STRUCTURE (THE ALKANE)



3-ethyl-1,1-dimethylcyclohexane

3 · FORMULA CHECK

Counted from the drawing: C10H20
One ring: CnH2n = C10H20 ✓

4 · BACK-CHECK

Returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane
■ Same as the given name
□ Different. Explanation: _____

✓ parent length: 6 C | ✓ positions: methyl on C1 (×2) | ✓ formula: C10H20 | □ back-check recorded

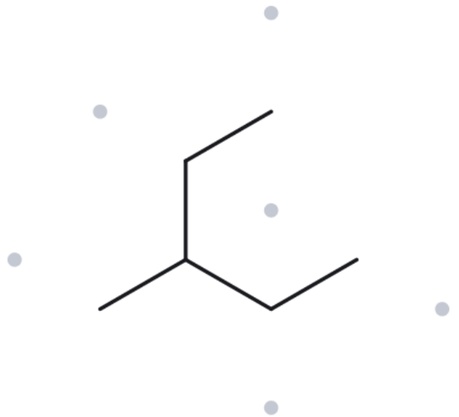
B10 CHALLENGE

Given name: 2-ethylbutane

1 · NAME DECODED

Parent: butane → 4-C chain (as written)
Substituents: ethyl on C2
Expected formula: 4 + 2 = 6 C; open chain → C6H14
Correct name: 3-methylpentane
Why it is wrong: Rule broken: the parent must be the longest continuous chain. The ethyl arm joins C2–C3–C4 to make a 5-C chain through C2, leaving a methyl on C3.

2 · BOND-LINE STRUCTURE (THE ALKANE)



2-ethylbutane

3 · FORMULA CHECK

Counted from the drawing: C6H14
Open chain: CnH2n+2 = C6H14 ✓

4 · BACK-CHECK

Returned by The Alkane: 3-methylpentane
□ Same as the given name
■ Different. Explanation: **Longest carbon chain has 5 carbons (pentane)**
Given name is deliberately wrong. Correct name: 3-methylpentane.

✓ parent length: 4 C | ✓ positions: ethyl on C2 | ✓ formula: C6H14 | □ back-check recorded